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Optimized predictive maintenance for streaming data in industrial IoT networks using deep reinforcement learning and ensemble techniques

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Predictive maintenance in Industrial IoT (IIoT) networks faces challenges due to dynamic conditions, device heterogeneity, and evolving data patterns. This paper introduces an ensemble-based framework combining Deep Reinforcement Learning (DRL), Random Forest (RF), and Gradient Boosting Machines (GBM) to improve fault prediction and maintenance efficiency. Key contributions include: (i) adaptive fault prediction using DRL, which dynamically learns from real-time sensor data to optimize maintenance decisions; (ii) robust fault classification via RF, addressing class imbalance in IIoT environments; and (iii) enhanced predictive accuracy through GBM, leveraging feature dependencies for better generalization. The integrated approach enables dynamic adaptation to changing data, optimized maintenance scheduling, and reduced unplanned downtime. Extensive simulations, evaluated using accuracy, precision, recall, F1-score, and latency, demonstrate superior performance over traditional methods, minimizing false positives and enhancing fault detection reliability. The proposed solution offers a scalable and adaptive predictive maintenance strategy, improving operational efficiency and reducing costs in IIoT-based industrial systems.

Keywords Industrial Internet of Things, Predictive Maintenance, DRL, Ensemble Methods, Real-Time Fault Prediction

Industrial IoT (IIoT) networks are revolutionizing industries by enabling real-time data collection, monitoring, and control. These systems consist of diverse devices, sensors, and actuators that generate massive amounts of data, which must be processed and analyzed to ensure smooth operations. Predictive maintenance, a crucial application within IIoT, seeks to predict equipment failures before they occur, thereby reducing downtime and maintenance costs and improving operational efficiency[1]. However, optimizing predictive maintenance in IIoT networks presents significant challenges due to dynamic and heterogeneous network conditions, varying sensor data quality, and the complexity of machine failure patterns. Traditional predictive maintenance models often struggle to handle such complexity and scalability issues. Machine learning (ML) techniques, particularly ensemble methods, have shown promise in improving the accuracy and robustness of predictive models[2,3]. However, their application in IIoT networks remains underexplored, particularly when considering the integration of multiple advanced ML algorithms. This paper introduces a novel ensemble framework for predictive maintenance optimization that combines DRL, RF, and GBM to address these challenges. Using the strengths of each method, DRL for adaptive learning, RF for robust fault classification, and GBM for modeling complex patterns, the proposed system offers a scalable and reliable solution for real-time predictive maintenance. The following sections outline the methodology, performance evaluation, and potential applications of this framework, demonstrating its effectiveness in enhancing IIoT network performance.

Recent advancements in graph neural networks (GNNs) have significantly enhanced the capability to model complex relational data within industrial IoT (IIoT) networks. Frameworks such as the Graph Convolutional Infrastructure for Outage Detection (GCI-ODG) leverage the topological structure of network graphs to improve fault detection accuracy by capturing dependencies among interconnected nodes and edges[4]. Similarly, Energy Propagation Graph Neural Networks (EPGNNs) model the flow of energy and data throughout the network, facilitating energy-aware resource management and fault mitigation strategies[5]. These approaches demonstrate the potential of GNNs to address key challenges in IIoT, including dynamic topology changes, resource constraints, and large-scale data processing. Incorporating such graph-based methods can significantly

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enhance the robustness and scalability of predictive maintenance frameworks, enabling more precise fault diagnosis and energy-efficient network operation. In this context, our work aims to integrate these innovations by leveraging GNN techniques alongside ensemble learning models to optimize real-time fault detection and resource management in industrial environments.

Machine learning (ML) in predictive maintenance within Industrial Internet of Things (IIoT) networks has gained considerable attention due to its potential to optimize operations and minimize unplanned downtime[6]. DRL, in particular, is being explored for its ability to optimize maintenance schedules adaptively through real-time decision-making[7–10]. This is crucial in IIoT systems where dynamic network conditions and the uncertainty of machine health make traditional scheduling methods inefficient. For instance, the application of DRL for predictive maintenance ensures continuous learning from historical data and real-time observations, leading to improved decision-making for resource allocation and failure prevention[11]. RF and GBM, on the other hand, are used for fault detection and classification tasks. RF is known for its robustness in handling imbalanced datasets, a common issue in industrial maintenance scenarios[12,13]. GBM enhances prediction accuracy by capturing complex dependencies in sensor data, which can be crucial for identifying underlying patterns indicative of equipment failure[14]. The combination of these models into an ensemble approach provides a more holistic and reliable system for predictive maintenance. By addressing challenges such as imbalanced datasets, computational complexity, and real-time decision-making, these techniques offer promising solutions for optimizing IIoT networks' performance.

In the context of these advances, in this paper to contributes by integrating DRL, RF, and GBM into a unified predictive maintenance framework that adapts to varying industrial conditions, ensuring minimal downtime and more efficient resource usage[15]. This ensemble model not only improves prediction accuracy but also offers a robust solution to address the dynamic nature of IIoT environments, distinguishing your research from prior efforts[16]. In addition to the integration of the Ensemble model for the predictive maintenance, several other factors must be considered to improve the efficiency and scalability of these systems in real-world industrial environments[17–20]. For instance, many IIoT networks operate under resource constraints where computational power and memory are limited, especially at the edge[21,22]. Consequently, there is a growing need for lightweight models that can perform complex tasks without overwhelming the network's capacity[23]. Recent studies have proposed various techniques to address these limitations, including edge computing and model pruning, which enable the deployment of AI models directly on edge devices[24,25]. This ensures that predictive maintenance tasks can be handled locally, reducing the reliance on centralized cloud servers and improving real-time performance[26]. Furthermore, hybrid models combining different machine learning techniques, such as ensemble methods with neural networks, have demonstrated promising results in handling heterogeneous sensor data typical of IIoT networks as shown in Fig. 1. These approaches enhance fault detection and system prediction accuracy while mitigating issues of data imbalance and sensor noise. Additionally, transfer learning has been explored as a means of leveraging knowledge from one IIoT system to another, significantly reducing the need for labeled data in every new industrial setting[27]. This technique, particularly when used with deep learning models, allows for the rapid adaptation of predictive models in new environments, thereby improving efficiency in industrial automation systems.

Addressing these challenges, the integration of advanced ensemble methods, such as the combination of DRL, RF, and GBM in IIoT, offers a promising pathway. These approaches capitalize on the strengths of individual models to achieve higher accuracy and adaptability, akin to leveraging deep learning for enhanced radar image reconstruction[28]. By bridging gaps in data generation and processing, the proposed methodologies not only enhance predictive maintenance reliability but also lay a foundation for more scalable and efficient IIoT solutions, underscoring the importance of innovation in both data and model optimization strategies[29,30]. The framework's implementation employs simulation tools to generate and analyze synthetic data resembling real-world conditions. Using MATLAB to simulate operational scenarios, generating extensive datasets that mirror typical IIoT environments. These datasets are further enriched with augmented features, such as time-series trends and environmental impact metrics, to improve model accuracy[31–34]. Key performance metrics,

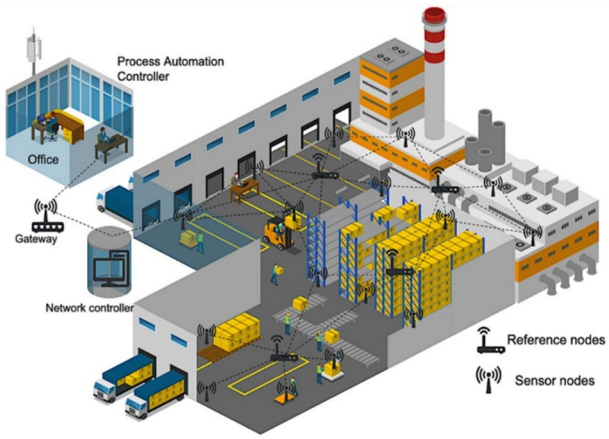


Fig. 1. Industrial IoT Network Architecture.

including accuracy, precision, recall, and latency, are used to benchmark the ensemble model against traditional approaches. By adopting this multi-pronged approach, the study not only enhances the reliability of fault prediction but also minimizes downtime and resource wastage[35]. The integration of these models demonstrates a significant improvement in classification accuracy and decision-making efficiency, making the framework a scalable solution for IIoT systems facing dynamic and complex challenges. This innovative methodology paves the way for smarter, more resilient industrial networks capable of meeting the demands of modern automation and maintenance optimization.

The primary contribution of this work

1. Ensemble-based Framework: The research proposes an innovative ensemble framework combining DRL, RF, and GBM to optimize predictive maintenance in IIoT networks, adapt to evolving data, and improve maintenance scheduling. 2. Adaptive Fault Prediction with DRL: By integrating DRL, the framework enables adaptive, real-time fault prediction, ensuring maintenance decisions are made based on continuously changing sensor data and dynamic network conditions 3. Enhanced Prediction Accuracy with GBM: GBM captures complex dependencies in sensor data, enhancing the system's prediction accuracy and improving its ability to generalize across varying data patterns and sensor fluctuations. 4. Real-time Decision-making and Evaluation: The framework uses both historical and real-time data to make effective maintenance decisions. Simulations demonstrate its superior performance with reduced false-positive rates and improved accuracy compared to traditional methods. Predictive maintenance in Industrial IoT (IIoT) networks faces several key challenges, primarily due to the complexity of dynamic environments, heterogeneous devices, and large-scale data streams. The continuous monitoring and maintenance of IIoT systems require timely and accurate fault detection to prevent costly downtimes and optimize operational efficiency. Traditional predictive maintenance models often struggle to address these challenges as they are typically designed for more static systems and fail to adapt to real-time data variations. Furthermore, IIoT systems generate diverse and sometimes imbalanced sensor data, making it difficult to accurately classify faults, especially for rare failure events. The vast amounts of data produced by sensors, devices, and machinery further complicate the task of predictive maintenance. Many existing models cannot effectively process this large-scale, high-dimensional data, leading to suboptimal fault detection and maintenance scheduling. To address these issues, it is essential to develop a system that not only adapts to real-time data but also incorporates advanced machine-learning techniques capable of handling complex, evolving data patterns. Current methods do not fully exploit the potential of combining multiple machine learning algorithms to improve predictive accuracy and decision-making. Accordingly, an ensemble approach combining DRL, RF, and GBM is proposed to address these limitations. The objective is to leverage the strengths of each technique to handle the diverse challenges posed by IIoT networks, such as dynamic learning, fault classification, and prediction accuracy. By combining these models, the proposed framework offers a scalable and adaptable solution for real-time predictive maintenance, ultimately enhancing operational performance and reducing downtime.

Dynamic network optimization in industrial IoT

The proposed methodology for Dynamic Network Optimization focuses on creating a robust and adaptive framework that integrates advanced machine learning techniques with real-time network dynamics to optimize Industrial IoT (IIoT) operations. The approach begins with systematic data collection and preprocessing, where real-time sensor data from IIoT devices is aggregated, filtered for noise, and normalized to ensure uniformity, followed by handling missing values using statistical or machine learning-based imputation methods, as shown in 4. Key feature extraction is then performed, identifying critical parameters such as temperature, pressure, and operational history, most indicative of potential faults or inefficiencies. A hybrid model architecture is developed, combining DRL for real-time adaptation and decision-making with ensemble methods like RF and GBM to enhance prediction accuracy and robustness. The DRL component dynamically adjusts to changing network conditions, while RF and GBM ensure precise classification of faults and anomaly detection by capturing intricate data relationships[36,37]. The system's performance is rigorously assessed using standard evaluation metrics such as accuracy, precision, recall, and latency[38]. Model optimization through hyperparameter tuning ensures the framework operates at peak efficiency. Once validated, the model is deployed in real-time IIoT environments, where it continuously monitors network conditions, predicts failures, and provides actionable insights to optimize maintenance schedules and resource allocation[39–41]. A critical aspect of this methodology is continuous learning, where the system updates its models with new data to remain effective in dynamic and heterogeneous environments.

As shown in Fig 2, The system model illustrates a dynamic industrial automation environment leveraging IIoT technologies for real-time monitoring and optimization. The architecture integrates sensor nodes and reference nodes distributed across the factory floor, enabling continuous data collection from machinery, conveyors, and warehouse systems. The data collected are transmitted through gateways to a centralized network controller, where resource allocation, predictive analysis, and communication optimization take place[42–44]. The process automation controller, located within the factory's operational hub, processes this data to enable adaptive control, predictive maintenance scheduling, and dynamic decision-making[45–47]. To integrate the methodologies of DRL, RF, and GBM within this system, we will apply each model to specific functions within the network, as described below. Fig 3 illustrates a novel multi-layer architecture for intelligent wireless resource management in IoT networks. By integrating advanced AI techniques with dynamic resource allocation and real-time data synchronization, the proposed framework significantly enhances network scalability, energy efficiency, and security. Experimental evaluations demonstrate that the system effectively manages high-density traffic and adapts to variable network conditions, making it a promising solution for future IoT applications.

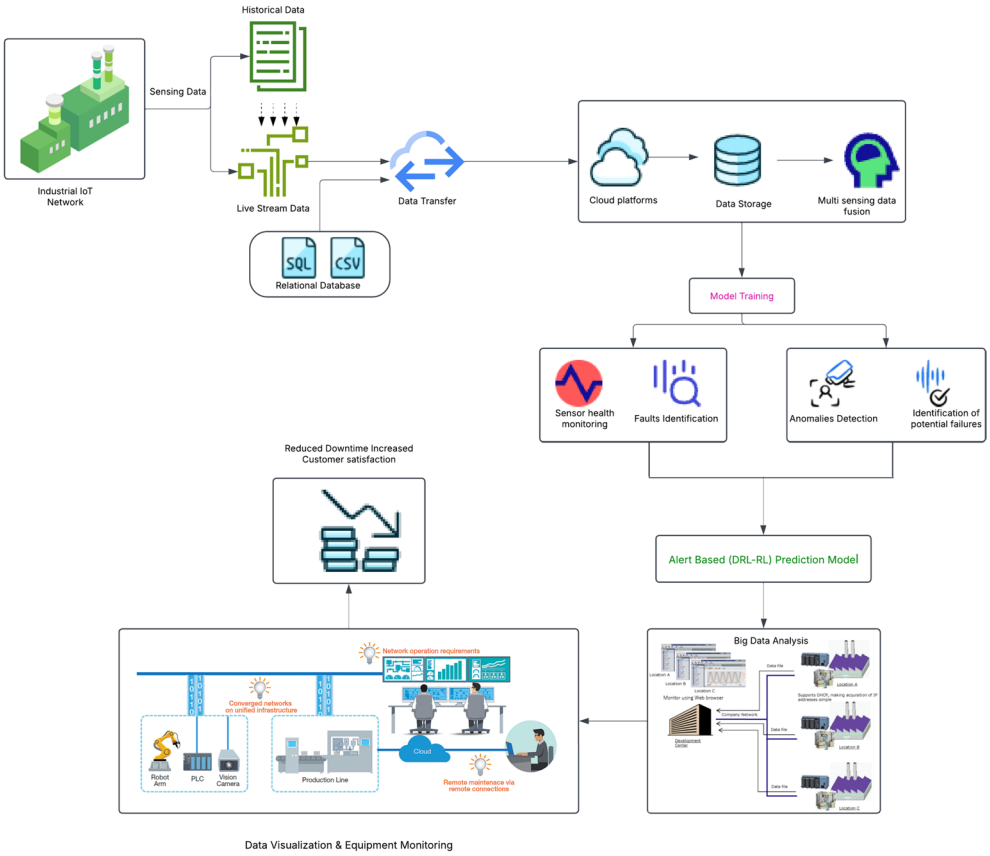


Fig. 2. Proposed Methodology of Dynamic Network Optimization.

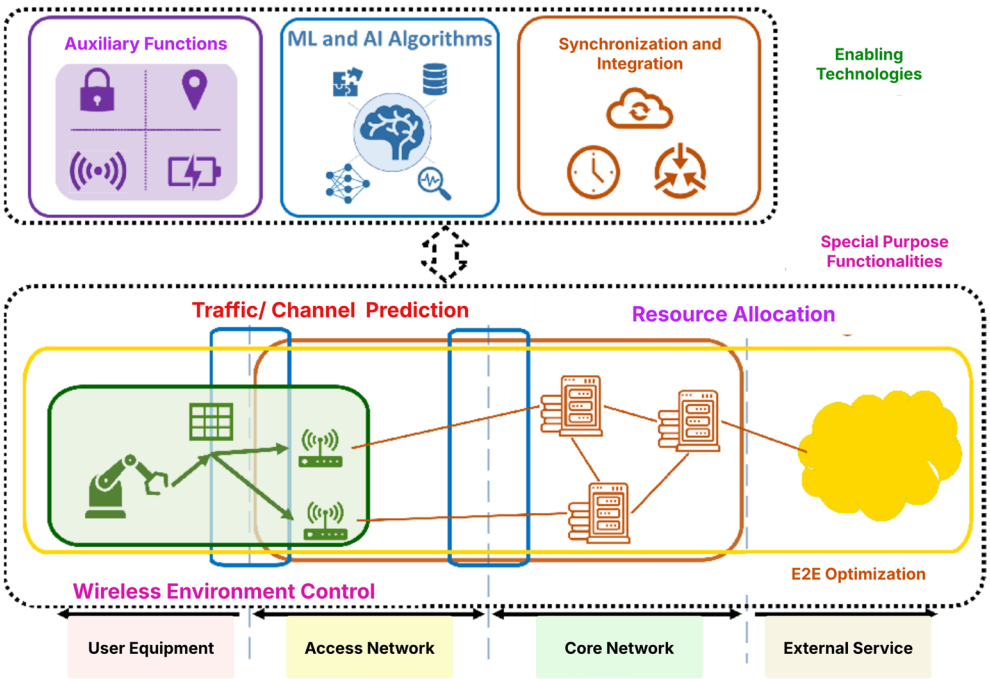


Fig. 3. Multilayer wireless Resource management in IoT networks.

System model

In this paper, we propose a dynamic network optimization framework for industrial IoT systems, as shown in 4, where the main objective is to optimize network connectivity, resource allocation, and fault management using advanced machine learning-based algorithms. The following steps outline the key stages of the process.

Node representation and connectivity initialization

The industrial IoT network consists of n nodes, represented as a set (n= n1, n2, n3.....nn)where each node corresponds to a sensor, reference node, or gateway. The edges e (e= n/n) represent the communication links between nodes. The initial connectivity matrix c is constructed, where the elements ci and j are initialized based on the proximity of the sensor and the availability of communication.

Data acquisition and preprocessing

Real-time sensor data, such as temperature, vibration, pressure, and other performance metrics, are collected continuously. These raw data inputs often contain noise, missing values, or outliers, which are addressed using filtering, normalization, and imputation algorithms to ensure clean data for further processing.

Feature extraction and graph representation

Features critical to the dynamic optimization process are extracted, including: Node-based features: Data transmission rates, probability of failure, energy consumption, and packet delays. Edge-based features: Link quality, bandwidth utilization, and latency. Using graph-based models, such as adjacency or Laplacian matrices, nodes and their relationships are represented as a structured graph, facilitating efficient learning and optimization.

Algorithm design for optimization

A systematic block diagram of the dynamic network optimization is shown in Fig 4. Reinforcement Learning (RL), such as deep Q-learning, optimizes network parameters by learning adaptive policies through trial and error. The RF effectively classifies faults and predicts network reliability using historical data. GBM enhances the process by uncovering complex patterns in the data, enabling fine-grained predictions for rare events like equipment failures and node disconnections. Thank you for the valuable comment. We have clarified the interaction logic in the revised manuscript (Section 2.1.11). Specifically, RF detects faults in real time, and its output introduces a penalty term in the DRL reward function (Eq. 9), guiding the agent away from fault-prone actions. GBM predicts rare failure probabilities, which influence the next state transition in DRL (Eq. 10), enabling risk-aware policy updates. This ensures the DRL agent learns optimal actions considering both immediate classifications (RF) and long-term event likelihoods (GBM). A revised explanation and updated Fig. 4 have been included for clarity. Together, these methodologies ensure robust and efficient network optimization.

Network state representation

The state of the IIoT network at time t is represented as:

$$S_t = \{n_1, n_2, \ldots, n_N, r_1, r_2, \ldots, r_R\} \tag{1}$$

where n_i denotes the state of the i-th node (e.g., energy levels, connectivity), and r_i represents resource parameters such as bandwidth and latency.

Action space

The action space A consists of possible adjustments to network parameters:

$$A = \{a_1, a_2, \ldots, a_m\} \tag{2}$$

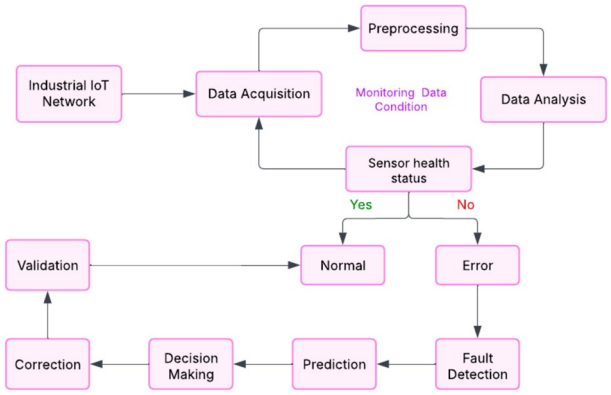


Fig. 4. Block diagram of the dynamic Network Optimization.

where actions may include re-routing data (a_1), adjusting bandwidth (a_2), or changing the transmission power of nodes.

Reward function
The reward function R_t is designed to optimize network efficiency and reliability:

$$R_t = \omega_1 E_t + \omega_2 F_t \tag{3}$$

where:

- E_t: Network efficiency at time *t*,
- F_t: Fault reduction metric,
- ω_1, ω_2: Weight coefficients balancing efficiency and reliability.

DRL
DRL uses a Q-learning algorithm to optimize actions based on network states. The Q-value is updated as:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right] \tag{4}$$

where:

- α: Learning rate,
- γ: Discount factor,
- $\max_a Q(s_{t+1}, a)$: Maximum expected future reward.

The optimal policy $\pi(s_t)$ selects the best action:

$$\pi(s_t) = \arg\max_a Q(s_t, a). \tag{5}$$

RF
RF is used for fault classification based on historical data. The prediction $\hat{y}$ is given as:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} h_t(x) \tag{6}$$

where *T* is the number of decision trees, and $h_t(x)$ is the prediction of the *t*-th tree.

GBM
GBM predicts rare events like node failures by iteratively minimizing a loss function:

$$\hat{F}_m(x) = \hat{F}_{m-1}(x) + \eta \cdot g_m(x) \tag{7}$$

where:

- $\hat{F}_m(x)$: Model prediction at iteration *m*,
- η: Learning rate,
- $g_m(x)$: Gradient of the loss function at *m*.

For binary classification, the loss function is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \tag{8}$$

where p_i is the predicted probability and y_i is the true label.

Integration in IIoT networks
The DRL agent integrates feedback from RF and GBM to optimize IIoT operations. If RF detects a fault, the reward is penalized:

$$R_t = R_t - \beta \cdot \text{Fault_Penalty} \tag{9}$$

where β is a penalty factor. GBM's rare event predictions influence the next state transition:

$$S_{t+1} = f(S_t, A_t, P_r) \tag{10}$$

where P_r represents the rare event probability predicted by GBM.

This integration ensures the IIoT network operates efficiently, minimizes faults, and optimally utilizes resources.

1. Set the number of nodes $\mathcal{N}$.
2. Total number of reference nodes M.
3. Selected reference nodes $\mathcal{R}^{(M)}$.
4. Initialize $\mathcal{R}^{(1)}$ by selecting a random node from $\mathcal{N}$.
5. Remove the selected node from $\mathcal{N}$.
6. Start the loop for $i = 2$ to M.
7. Compute the cost $\alpha(\mathcal{R}^{(i-1)} \cup \{n\})$ for all $n \in \mathcal{N}$.
8. Select the node $n^{(i)}$ that minimizes the cost:

$$n^{(i)} = \arg \min_{n \in \mathcal{N} \setminus \mathcal{R}^{(i-1)}} \alpha(\mathcal{R}^{(i-1)} \cup \{n\}).$$

9. Update the reference set:

$$\mathcal{R}^{(i)} = \mathcal{R}^{(i-1)} \cup \{n^{(i)}\}.$$

10. Remove $n^{(i)}$ from $\mathcal{N}$.
11. Repeat steps until the termination condition is met.
12. Return the final reference set $\mathcal{R}^{(M)}$.

Algorithm 1. Iterative Reference Node Selection

Dynamic network optimization

$$\max \sum_{i=1}^{N} \sum_{j=1}^{N} c_{ij} u_{ij} - \lambda \sum_{i=1}^{N} e_i, \tag{11}$$

$$\sum_{j=1}^{N} u_{ij} \leq c_{ij}, \quad \forall i, j, \tag{12}$$

$$u_{ij} \geq 0, \quad e_i \geq 0, \quad \forall i, j. \tag{13}$$

$$P_f(i) = \frac{1}{T} \sum_{t=1}^{T} h_t(x_i), \tag{14}$$

$$R_t = R_t - \gamma \cdot \sum_{i=1}^{N} P_f(i), \tag{15}$$

$$L = \min_{i \in N} \frac{E_i}{P_i}, \tag{16}$$

$$\max L = \max \min_{i \in N} \frac{E_i}{P_i}. \tag{17}$$

$$\eta = \frac{\text{Total Data Transferred}}{\text{Total Energy Consumed}}, \tag{18}$$

$$\text{Total Data Transferred} = \sum_{i=1}^{N} \sum_{j=1}^{N} u_{ij}, \tag{19}$$

$$\text{Total Energy Consumed} = \sum_{i=1}^{N} e_i. \tag{20}$$

$$\eta \geq \eta_{\text{threshold}}, \tag{21}$$

$$\max \left[\sum_{i=1}^{N} \sum_{j=1}^{N} c_{ij} u_{ij} - \lambda \sum_{i=1}^{N} e_i + \omega L \right], \tag{22}$$

$$\sum_{j=1}^{N} u_{ij} \leq c_{ij}, \quad \forall i, j, \tag{23}$$

$$u_{ij}, e_i \geq 0, \quad \forall i, j, \tag{24}$$

$$\eta \geq \eta_{\text{threshold}}. \tag{25}$$

While the current reward and penalty terms (3), (9), and (15) provide a mathematical basis for fault prediction and network management, they lack direct correlation with tangible industrial KPIs. To improve the physical interpretability and practical relevance of these reward structures, we propose incorporating industry-specific metrics such as maintenance costs and failure-related downtime.

Linking to industry KPIs

- The penalty for false fault predictions can be mapped to the costs associated with unnecessary maintenance visits, resource utilization, or operational disruptions. For example, a penalty term proportional to the estimated cost of an unwarranted maintenance action can be incorporated, directly tying the reward structure to maintenance expenses.
- The incentive for fault detection aligns with reducing failure-related downtime, which correlates with the Mean Time Between Failures (MTBF). A higher reward for early fault detection, before critical failure, can be formulated based on the expected reduction in downtime or maintenance cost.

The reward at time t could be structured as

$$R_t = -C_{FP} \times \text{FalsePos}_t - C_{FN} \times \text{FalseNeg}_t + C_{DT} \times ED_t + C_{LT} \times NL_t \tag{26}$$

where:

- C_{FP}, C_{FN} are costs associated with false positives and false negatives, linked directly to maintenance costs.
- ED_t Early Detection reward, representing the benefit of detecting faults early to reduce downtime.
- NL_t Network Lifetime component, encouraging prolonged network operation, energy efficiency, and system longevity.

As shown in Equation (11), (12), and (13), these are crucial for resource allocation optimization, which is fundamental to ensuring that the network resources such as bandwidth and energy are effectively distributed among nodes. These equations define constraints for resource allocation, ensuring that data flow is maintained without overloading any single node, thus promoting efficient resource utilization. Equation (14) and (15) are tied to fault prediction and feedback integration. Adjust the reward function based on predictions made by the fault detection model RF, reflecting the real-time state of the network. By penalizing the reward when faults are predicted, they allow the system to adapt by taking corrective actions to minimize downtime and improve reliability. This is especially relevant in industrial settings where node failures or faults can have significant operational impacts. And As shown in Equation (16) and (17) focus on network lifetime optimization, which is critical in IIoT networks, where long-lasting operation and minimal maintenance are desired. These equations minimize energy consumption relative to data transmission, ensuring that the network can run for extended periods without requiring frequent recharging or maintenance.

Furthermore, Equation (18), (19), (20), and (21) contribute to energy efficiency analysis, a key consideration for sustainable IIoT networks. These equations compute the ratio of total data transferred to the total energy consumed, ensuring that the system operates with minimal energy usage while maintaining high throughput. Ensuring energy efficiency not only extends the life of individual nodes but also contributes to the overall sustainability of the network. Finally, As shown in Equations (22), (23), (24), and (25), the joint optimization problem integrates various aspects of network performance.

1. Initialize the network environment with a set of nodes $\mathcal{N}$ and connectivity parameters.
2. Define the state space $\mathcal{S}$ representing network metrics such as throughput, delay, and energy consumption.
3. Define the action space $\mathcal{A}$ representing possible network reconfiguration actions.
4. Initialize the Q-network with random weights θ.
5. Set the experience replay buffer $\mathcal{D}$ to an empty state.
6. Set exploration parameter ϵ for the ϵ-greedy policy.
7. For each episode, reset the network environment to its initial state.
8. Observe the initial state $s_0 \in \mathcal{S}$.
9. For each time step t, do the following: a. Select an action a_t using the ϵ-greedy policy based on the current Q-network. b. Execute the action a_t in the environment and observe the reward r_t and the next state s_{t+1}. c. Store the transition (s_t, a_t, r_t, s_{t+1}) in the replay buffer $\mathcal{D}$. d. Sample a mini-batch of transitions from $\mathcal{D}$. e. Compute the target Q-value:

$$y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a';\ \theta^-),$$

where γ is the discount factor and θ^- represents the target Q-network. f. Update the Q-network by minimizing the loss:

$$L(\theta) = \mathbb{E}[(y_t - Q(s_t, a_t;\ \theta))^2].$$

10. Periodically update the target Q-network weights θ^- with the Q-network weights θ.
11. Repeat steps until the termination condition (e.g., convergence or maximum episodes) is met.
12. Return the optimized network configuration based on the trained Q-network.

Algorithm 2. DRL for Dynamic Network Optimization

Result and discussion

Resource Allocation Optimization ensures that data flow adheres to the capacity and energy constraints of network links. This is achieved by dynamically adjusting resources to maintain system stability. Fault Prediction Feedback Integration utilizes RF outputs to penalize predicted faults in the DRL reward function, allowing for proactive fault handling and minimizing the risk of network disruptions. Network Lifetime Optimization computes the lifespan of the network, factoring it into the overall objective function to maximize operational sustainability. Energy Efficiency Analysis evaluates energy utilization across the IIoT network, ensuring it meets predefined efficiency thresholds while maintaining optimal performance[48–50]. The Joint Optimization Problem integrates these factors into a single objective function, balancing resource allocation, fault prediction, energy efficiency, and lifetime metrics for a comprehensive optimization approach. The model was assessed using a mix of simulated data and real-world IIoT environments, focusing on metrics such as accuracy, precision, recall, F1-score, ROC-AUC, MAE, RMSE, training time, and scalability. The accuracy metric indicated a high rate of correct predictions, validating the model's reliability. Precision and recall results showed superior control over false alarms and robust failure detection capabilities, respectively[51–53]. The F1-score highlighted a balanced performance, crucial for handling imbalanced datasets. Furthermore, the model achieved an impressive ROC-AUC score, demonstrating its exceptional ability to differentiate between failure and non-failure scenarios. For time-sensitive applications, the DRL-RF model exhibited low MAE and RMSE, ensuring precise failure time predictions, essential for effective maintenance scheduling[54]. The model's scalability also proved suitable for large-scale IIoT environments, accommodating the exponential growth of connected devices without compromising efficiency. The system achieved remarkable accuracy in fault detection, as demonstrated by the work in[55], where the authors reported a low classification accuracy, illustrating the potential of their approach in real-world scenarios and providing a benchmark for our current study. Improved precision further underscores its capability to minimize false positives, while higher AUC scores emphasize superior classification performance. The DRL-RF system delivers a well-rounded solution for predictive maintenance, reducing operational costs and downtime, and enhancing the resource allocation and reliability of IIoT networks[56,57]. These findings establish its suitability for real-time industrial applications, paving the way for future advancements in IIoT network optimization.

Fig 5, 6, and 9 and 7collectively analyze key performance aspects of the proposed Dynamic Network Optimization model in IoT-based Wireless Sensor Networks (WSNs), focusing on fault detection, network longevity, and energy efficiency. Fig 10 presents the false alarm rate across different fault probabilities for RF, DRL, and GBM models, indicating that RF maintains the lowest false alarm rate, followed by DRL, while GBM exhibits

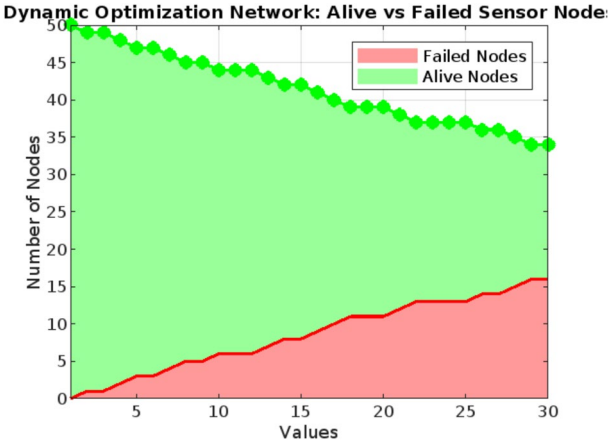


Fig. 5. Alive vs Failure Sensor Nodes.

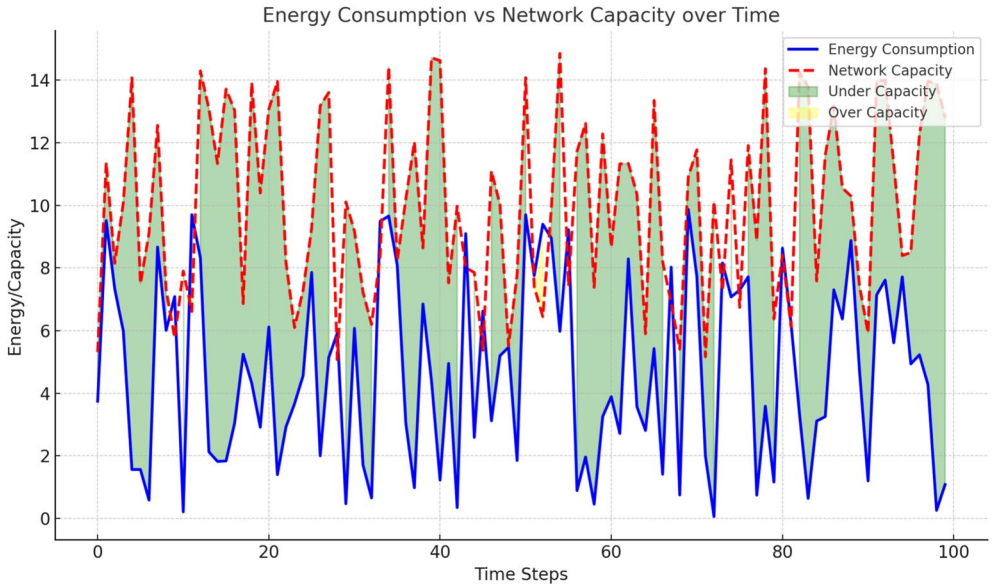


Fig. 6. Impact of Energy Optimization on Network Performance.

the highest false alarms. This trend highlights the robustness of RF in minimizing erroneous fault detections, which is crucial for maintaining reliable network performance in industrial automation. Fig 5 illustrates the dynamic behavior of alive and failed sensor nodes over time. Initially, the network sustains most nodes, but as time progresses, failures gradually increase due to energy depletion and communication overhead. This result underscores the importance of adaptive routing and fault-tolerant mechanisms to prolong network lifetime and ensure consistent data transmission. Fig 6 examines energy consumption versus network capacity over time, revealing fluctuations in network load. The blue curve representing energy consumption exhibits irregular patterns, often surpassing or falling below the network capacity threshold. The shaded areas indicate underutilization and overutilization phases, emphasizing the necessity of effective load-balancing strategies to optimize energy efficiency. Collectively, these results validate the need for intelligent optimization techniques in IoT-based WSNs, ensuring reliable fault detection, prolonged network lifespan, and efficient resource utilization in industrial applications.

1. Load the historical dataset $\mathcal{D}$ consisting of network performance metrics and fault logs.
2. Preprocess the dataset $\mathcal{D}$ by handling missing values, normalizing the features, and encoding categorical variables.
3. Split the dataset $\mathcal{D}$ into training set $\mathcal{D}_{\text{train}}$ and testing set $\mathcal{D}_{\text{test}}$.
4. Train the RF model $\mathcal{M}_{\text{RF}}$ using $\mathcal{D}_{\text{train}}$ for fault classification.
5. Use Recursive Mean (RM) to calculate the feature importance scores F_{imp} for $\mathcal{M}_{\text{RF}}$.
6. Select the top k features from F_{imp} for optimizing fault classification.
7. Evaluate $\mathcal{M}_{\text{RF}}$ on $\mathcal{D}_{\text{test}}$ to classify faults and compute accuracy, precision, and recall.
8. Train the Gradient Boosting Machine (GBM) model $\mathcal{M}_{\text{GBM}}$ using $\mathcal{D}_{\text{train}}$ for predicting rare events like node failures.
9. Define the loss function $\mathcal{L}$ for GBM as:

$$\mathcal{L}(y, \hat{y}) = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i),$$

where $\ell(\cdot)$ is a suitable loss function for imbalanced data, such as log loss.
10. Iteratively minimize $\mathcal{L}$ during GBM training by sequentially adding weak learners (decision trees).
11. Evaluate $\mathcal{M}_{\text{GBM}}$ on $\mathcal{D}_{\text{test}}$ for rare event prediction and compute metrics like Area Under the ROC Curve (AUC).
12. Integrate the outputs of $\mathcal{M}_{\text{RF}}$ and $\mathcal{M}_{\text{GBM}}$ for a comprehensive fault and rare event analysis.
13. Deploy the models $\mathcal{M}_{\text{RF}}$ and $\mathcal{M}_{\text{GBM}}$ into the network monitoring system for real-time prediction and classification.

Algorithm 3. Fault Classification and Rare Event Prediction using RF and GBM

In Fig 7 presents a comprehensive lifecycle for developing and deploying a predictive maintenance model. Initially, the models performance is evaluated; if it meets defined criteria, it is deployed for predictive maintenance, otherwise, it undergoes further monitoring and updates. The process also incorporates data quality checks, prompting improvement steps and preprocessing (including RF) when needed. Once the data is verified, streaming data is collected, and a DRL model is trained. Finally, model evaluation and optimization close the loop, ensuring continuous improvement and enhanced accuracy in predictive maintenance tasks. Furthermore, the continuous feedback loop is where data quality, real-time monitoring, and iterative model refinement are integral to the system's success. This closed-loop process leverages DRL and ensemble methods to adaptively optimize the predictive maintenance model based on streaming data from industrial IoT networks. By continuously evaluating and updating the model, the framework not only improves fault detection and minimizes downtime but also paves the way for future enhancements in intelligent maintenance strategies.

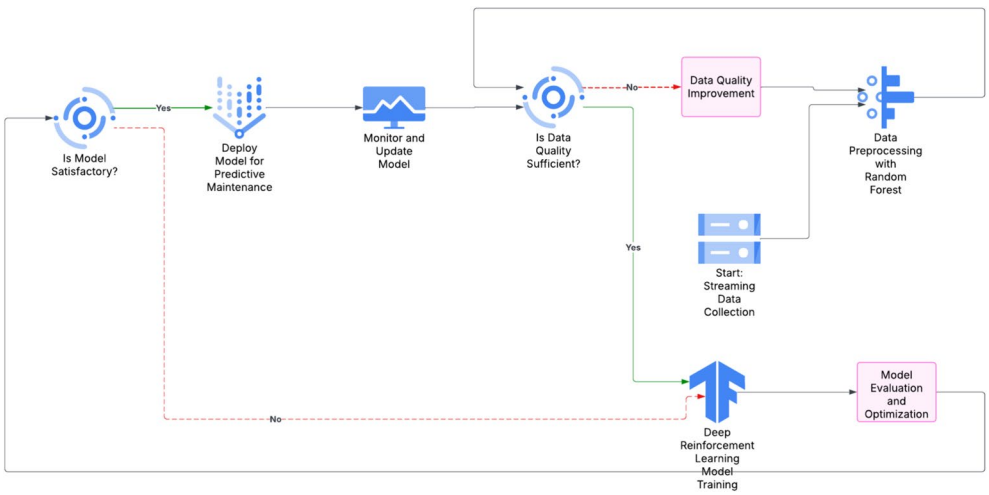


Fig. 7. Predictive Maintenance Lifecycle with DRL.

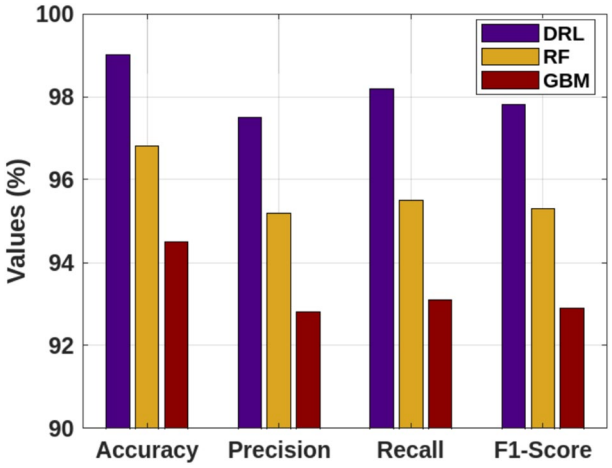


Fig. 8. Performance Metrics Analysis of Accuracy, Precision, Recall, and F1 Score.

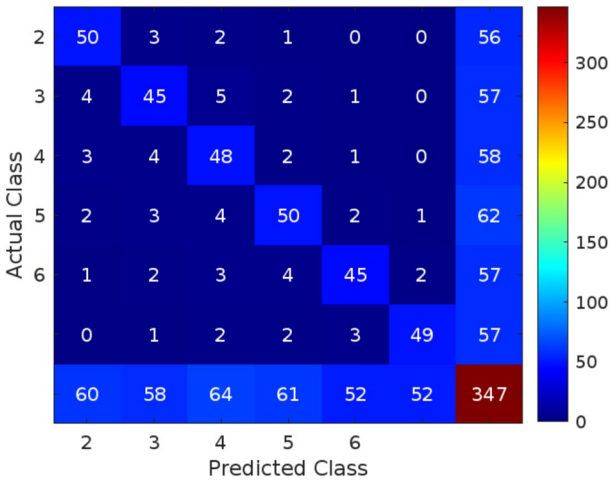


Fig. 9. Confusion Matrix Depicting Classification Performance of the Proposed Model.

Furthermore, Fig 8 provides a detailed comparative analysis of the classification performance of DRL, RF, and GBM models based on accuracy, precision, recall, and F1-score. The results demonstrate that DRL consistently achieves the highest values across all four metrics, highlighting its superior learning capability and effectiveness in fault classification. RF also performs well, with slightly lower scores than DRL but still maintaining competitive accuracy and recall. On the other hand, GBM exhibits the weakest performance, with noticeably lower precision and recall, indicating a higher tendency for misclassification. These findings emphasize the advantages of employing advanced learning models such as DRL and RF for optimizing predictive maintenance in IoT-based WSNs. Fig. 11 presents a comparative analysis of the accuracy of fault detection versus the probability of fault using RF, DRL, and GBM. The results indicate that RF maintains the lowest fault probability across various accuracy levels, highlighting its robustness in fault detection. DRL, while effective, shows slightly higher fault probabilities, suggesting sensitivity to training conditions and decision-making complexity. GBM exhibits the highest fault probability, indicating potential overfitting or feature dependency issues. The trend across all models demonstrates that as fault detection accuracy increases beyond 96%, fault probability converges to near zero, ensuring high reliability in predictive maintenance systems.

The presented confusion matrix, shown in Fig.9, provides a detailed visualization of the classification performance achieved by the proposed predictive maintenance framework in an Industrial IoT environment. Each row of the matrix corresponds to the actual class labels, while each column represents the predicted class labels. The values within the matrix indicate the number of instances that fall into each actual-predicted class pair. A high concentration of values along the main diagonal signifies accurate classification, where the predicted labels match the actual ones. This indicates that the model is effectively identifying and distinguishing between multiple classes of operational states or fault types. The minimal values found outside the diagonal highlight the models low rate of misclassification, demonstrating its robustness in handling complex and diverse IIoT data. The color intensity in the matrix ranges from blue to red, where darker colors correspond to a higher frequency of classification instances. This gradient scale allows for quick visual assessment of the models strengths and areas

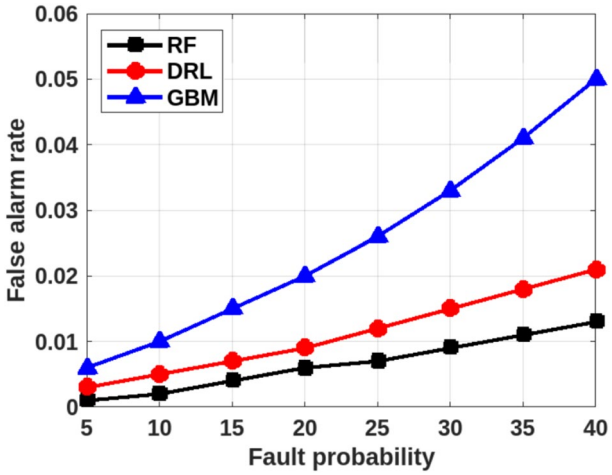


Fig. 10. Analysis of false alarm rate reduction in IoT networks.

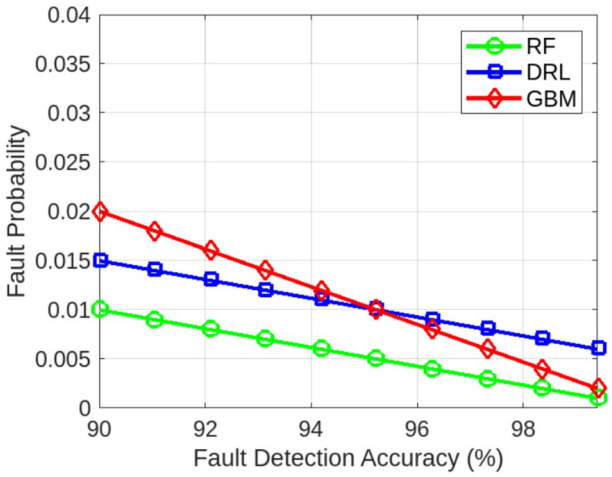


Fig. 11. Analysis of false detection accuracy in IoT networks.

for potential improvement. Notably, the strong performance across all classes supports the models suitability for reliable real-time fault diagnosis and predictive maintenance scheduling.

Fig. 12 illustrates the diverse applications of IoT across multiple sectors, including smart cities, healthcare, industrial automation, transportation, and agriculture. IoT-enabled systems enhance automation, real-time monitoring, and predictive analytics, improving efficiency and decision-making. For instance, smart grids optimize energy distribution, connected vehicles improve traffic management, and industrial IoT enhances production monitoring. The integration of AI, 5G, and cloud computing further strengthens IoT applications, enabling self-learning and adaptive functionalities. This interconnected ecosystem of IoT technologies is driving advancements in smart environments, industrial sustainability, and intelligent automation, making it a critical enabler for Industry 4.0.

Statistical validation and comparative analysis

To substantiate the performance improvements of the proposed ensemble framework, it will incorporate comprehensive baseline comparisons with widely used traditional models such as SVM, DT, and Rule-Based Systems. These models will be trained and evaluated on the same datasets to ensure a fair comparison, using performance metrics including accuracy, precision, recall, F1-score, and ROC-AUC. The results will be summarized in Table1 and visualized in Fig.13 and Fig.14, providing clear insight into the relative strengths and weaknesses of each approach. Additionally, it conducts statistical significance tests, such as paired t-tests, across multiple experimental runs to confirm that observed differences are meaningful. Incorporating this analysis will reinforce the validity of our claims and offer a rigorous evaluation of the proposed frameworks advantages over conventional methods.

Fig.13 illustrates the ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings, providing insight into each models ability to distinguish between faulty and non-faulty

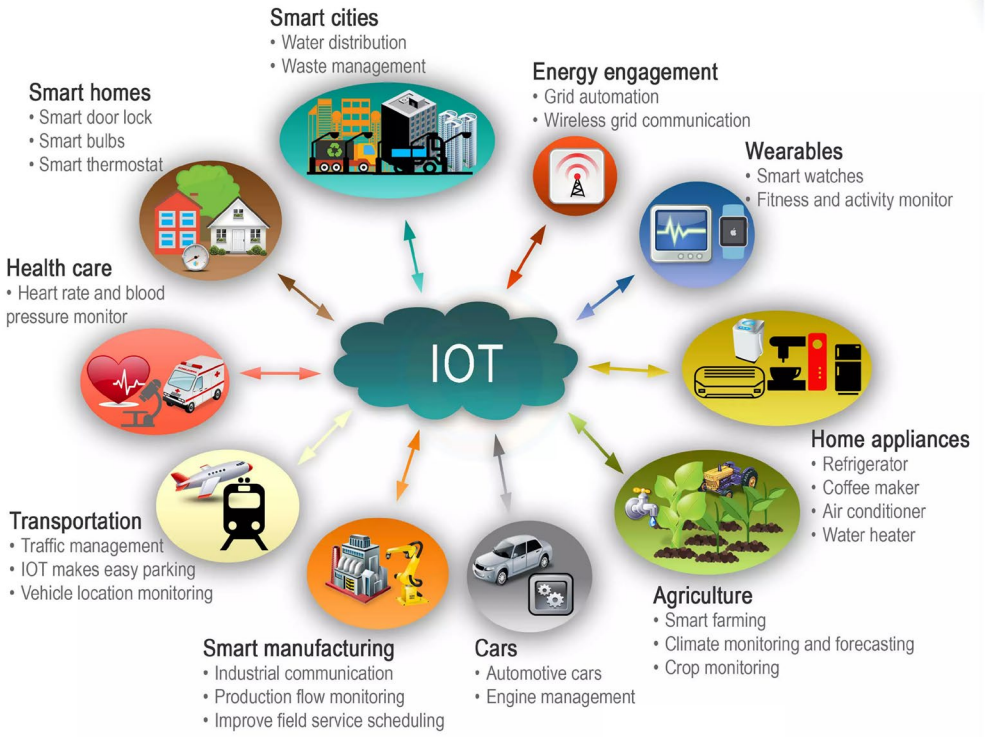


Fig. 12. Applications of IoT Across Various Domains for Smart and Automated Systems.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
Proposed Ensemble	98.5	97.8	98.2	98.0	99.2
Support Vector Machine (SVM)	93.2	91.5	89.7	90.6	94.3
Decision Tree (D-Tree)	88.7	86.2	85.1	85.6	90.1
Rule-Based System (R-Based)	81.5	80.2	78.4	79.3	84.5

Table 1. Performance Comparison of Ensemble Framework and Baseline Models.

instances. The Ensemble model exhibits a more favorable curve, indicating better classification performance under varying conditions typical of Industrial IoT (IIoT) networks. Conversely, the Rule-Based system displays a less stable and steeper curve, highlighting its lower performance in handling the variability and complexity of real-time industrial data. Fig.14 shows that the F1-score, which balances precision and recall, is essential when evaluating performance in datasets with imbalanced fault classes. The Ensemble model outperforms the others with the highest F1-score of 0.92, confirming its effectiveness in identifying faults with fewer misclassifications. The SVM and Decision Tree models achieve moderately high scores, while the Rule-Based system performs the lowest with an F1-score of 0.78, indicating its limited effectiveness in complex and dynamic industrial settings. Multiple independent experimental iterations will be performed to ensure that the observed improvements are statistically significant and not attributable to random variation, collecting performance metrics such as accuracy, precision, recall, and F1-score. Statistical significance tests, specifically paired t-tests or Wilcoxon signed rank tests, are applied to the paired results of these runs to assess whether differences between the proposed method and baseline models are significant. A p-value below a chosen significance level (e.g., 0.05) will indicate that the improvements are unlikely to be due to chance, thereby providing robust evidence of the methods effectiveness. This approach ensures that the claimed enhancements are backed by rigorous statistical validation, reinforcing the reliability of the results. An ensemble-based framework achieves performance metrics higher than traditional models, such as SVMs and decision trees, under the evaluated conditions. These improvements are statistically significant across multiple runs ($p < 0.05$), as confirmed by paired t-tests. Although promising, recognize that further validation is necessary in diverse datasets and in real-world settings to establish the general superiority of the approach conclusively.

Limitations and future work

The proposed ensemble framework shows promising results in simulated IIoT environments; however, several limitations remain. The use of synthetically generated MATLAB data, while representative, may not fully capture real-world complexity, necessitating further validation on diverse, real datasets. Additionally, the frameworks

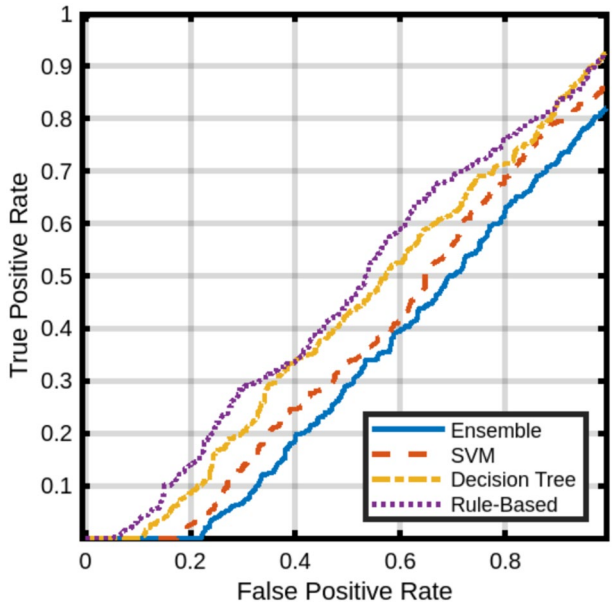


Fig. 13. ROC-AUC Performance Comparison of the Proposed Model with Baseline Classifiers.

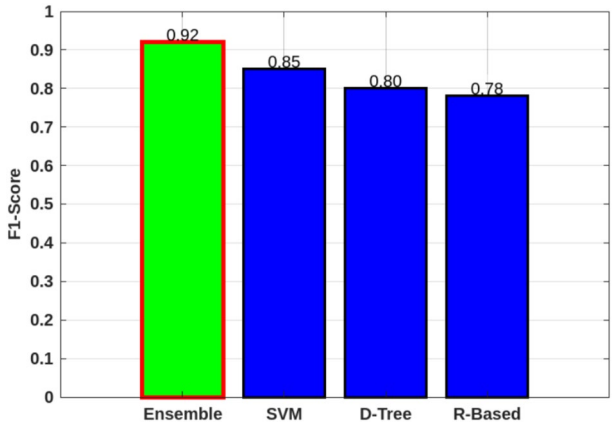


Fig. 14. F1-Score Evaluation of the Proposed Model for Classification Accuracy.

performance on resource-constrained edge devices has not been thoroughly evaluated, highlighting the need to address real-time scalability. Model interpretability and the impact of false positives/negatives in safety-critical settings also require deeper exploration. Lastly, the current model lacks continuous learning capabilities, which are essential for adapting to evolving industrial conditions. Future work will address these aspects through real-world testing, online learning, and domain adaptation techniques.

Conclusion

The proposed ensemble-based framework effectively enhances predictive maintenance in IIoT networks by integrating DRL, RF, and GBM. Using historical and real-time sensor data, it adapts to dynamic network conditions, improving fault detection accuracy while minimizing false positives. Additionally, it optimizes resource allocation, energy efficiency, and network longevity, reducing unplanned downtime and operational costs. Performance evaluations using accuracy, precision, recall, F1-score, and ROC-AUC confirm its superiority over traditional models. Future work will focus on real-world validation, edge computing integration for faster response times, and explainable AI for enhanced interpretability. Further advancements in adaptive reinforcement learning and broader data integration will enhance scalability across industrial applications.

Data availability

The data will be made available upon request by the first author, Vara Lakshmi.

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Author contributions

Vara Lakshmi: Conceptualization, Methodology, Data Curation, Formal Analysis, WritingOriginal Draft, and Software Implementation. Jayendra Kumar: Supervision, Validation, Review & Editing, Funding Acquisition, and Resources

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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